

Friday, Apr 5

Censoring Specification

Without much loss of generality we will limit the discussion here to right-censoring. We define an indicator variable D_i such that

$$D_i = \begin{cases} 1, & \text{if the } i\text{-th observation is not censored,} \\ 0, & \text{if the } i\text{-th observation is censored.} \end{cases}$$

The variable D_i can be viewed as another response variable which depends on the actual time to event, T_i , as well as whatever is responsible for the censoring. In what follows we will let t_i and d_i denote observed values of T_i and D_i respectively.

Let t_i be the actual time-till-event if $d_i = 1$, and the *lower-bound* on the time-till-event if $d_i = 0$ so that the *actual* time-till-event is *greater than or equal to* t_i . Under certain assumptions about how censoring occurs, the likelihood function is

$$L = \prod_{i=1}^n f(t_i)^{d_i} P(T_i \geq t_i)^{1-d_i}.$$

where $f(t_i)$ is the *probability density function* of T_i , and $P(T_i \geq t_i)$ is the probability that T_i is *at least* t_i (this is also called the *survival function*). Note that

$$f(t_i)^{d_i} P(T_i \geq t_i)^{1-d_i} = \begin{cases} f(t_i), & \text{if } d_i = 1 \text{ (i.e., not censored),} \\ P(T_i \geq t_i), & \text{if } d_i = 0 \text{ (i.e., censored),} \end{cases}$$

so the indicator variable d_i simply selects the appropriate term for computing the likelihood of an observation depending on whether or not it was censored.

Specification of Right-Censoring in Surv

For *right-censoring*, the response variable can be specified as `Surv(t,d)` where `t` is (a) the actual time to event if there is no censoring or (b) the lower bound on time to event if the observation is right-censored, and `d` is either an indicator variable (i.e., 0 or 1) or a logical variable (i.e., FALSE or TRUE) where we have `d = 1` or `d = TRUE` if the observation is *not* censored.

Example: Consider an AFT model for the leukemia data.

```
library(survival) # for leukemia and survreg
head(leukemia) # status=1 if remission ended at that time, status=0 if right-censored
```

	time	status	x
1	9	1	Maintained
2	13	1	Maintained
3	13	0	Maintained
4	18	1	Maintained
5	23	1	Maintained
6	28	0	Maintained

```
m <- survreg(Surv(time, status) ~ x, dist = "lognormal", data = leukemia)
summary(m)$table
```

	Value	Std. Error	z	p
(Intercept)	3.5789	0.2847	12.5712	3.039e-36
xNonmaintained	-0.7245	0.3803	-1.9049	5.679e-02
Log(scale)	-0.1454	0.1695	-0.8576	3.911e-01

Alternatively suppose we had a variable `censored` that told us if the observation was censored or not.

```
leukemia$censored <- factor(leukemia$status, labels = c("yes", "no"))
head(leukemia)
```

	time	status	x	censored
1	9	1	Maintained	no
2	13	1	Maintained	no
3	13	0	Maintained	yes
4	18	1	Maintained	no
5	23	1	Maintained	no
6	28	0	Maintained	yes

Then we specify the censoring as follows.

```
m <- survreg(Surv(time, censored == "no") ~ x, dist = "lognormal", data = leukemia)
summary(m)$table
```

	Value	Std. Error	z	p
(Intercept)	3.5789	0.2847	12.5712	3.039e-36
xNonmaintained	-0.7245	0.3803	-1.9049	5.679e-02
Log(scale)	-0.1454	0.1695	-0.8576	3.911e-01

It is useful to note that we can see how `Surv` codes the response variable for censoring. This is useful if you want to verify that you have used `Surv` correctly.

```
leukemia$ysurv <- Surv(leukemia$time, leukemia$censored == "no")
head(leukemia)
```

	time	status	x	censored	ysurv
1	9	1	Maintained	no	9
2	13	1	Maintained	no	13
3	13	0	Maintained	yes	13+
4	18	1	Maintained	no	18
5	23	1	Maintained	no	23
6	28	0	Maintained	yes	28+

As before, interpretation is facilitated by applying the exponential function to the parameter estimates.

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	35.8327	20.51	62.604
xNonmaintained	0.4846	0.23	1.021

```
leukemia$x <- relevel(leukemia$x, ref = "Nonmaintained")
m <- survreg(Surv(time, status) ~ x, dist = "lognormal", data = leukemia)
summary(m)$table
```

	Value	Std. Error	z	p
(Intercept)	2.8544	0.2539	11.2415	2.550e-29
xMaintained	0.7245	0.3803	1.9049	5.679e-02
Log(scale)	-0.1454	0.1695	-0.8576	3.911e-01

```
exp(cbind(coef(m), confint(m)))
```

```
                2.5 % 97.5 %  
(Intercept) 17.364 10.5563 28.561  
xMaintained  2.064  0.9793  4.349
```

Interval-Censoring

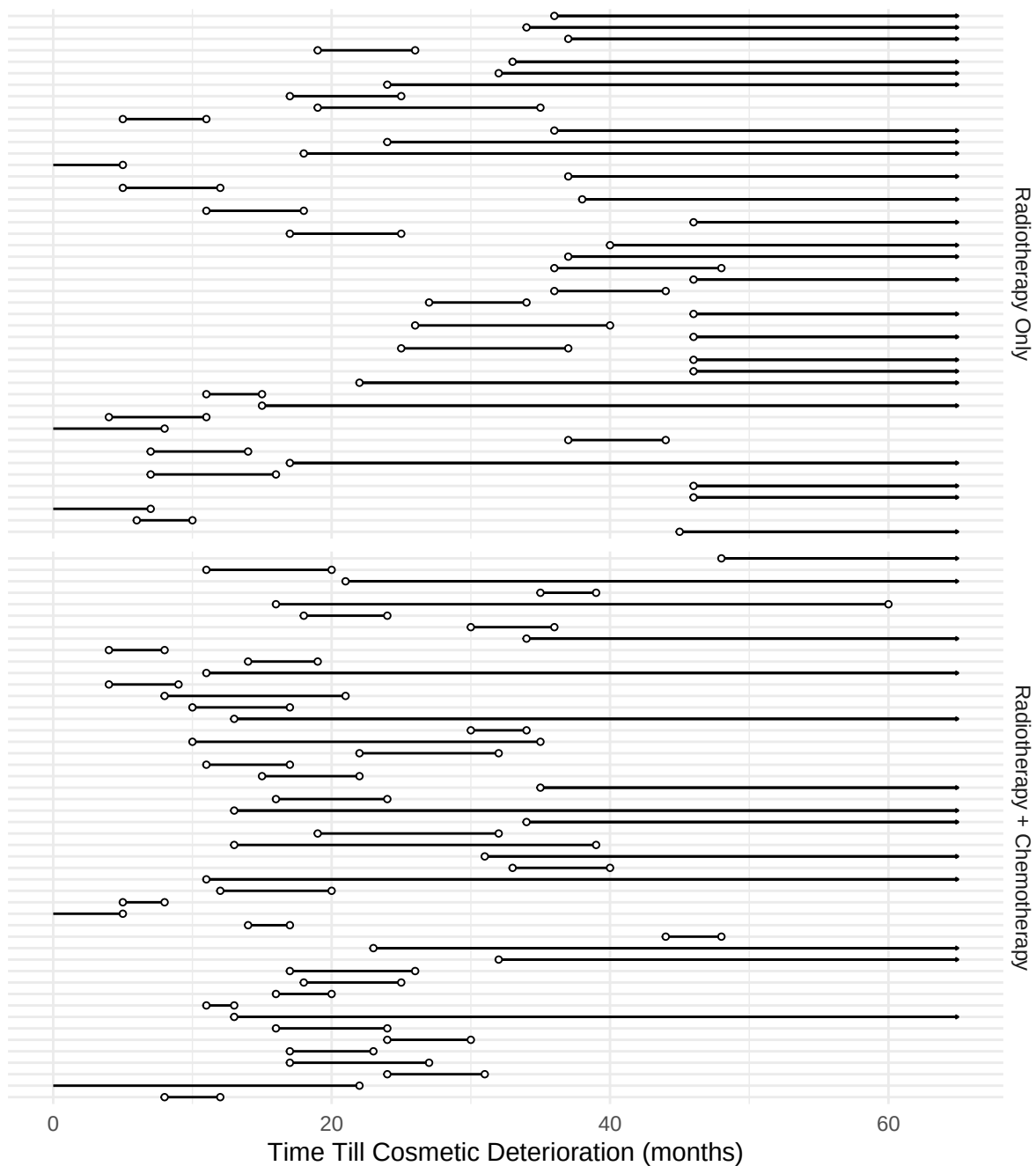
Interval censoring occurs when T_i is only known to be between two numbers such that $a < T_i < b$ where $0 \leq a < b \leq \infty$. Note that right-censoring is a special case where $b = \infty$, and left-censoring is a special case where $a = 0$.

Example: Consider the following data from a study of the time till cosmetic deterioration for breast cancer patients undergoing radiotherapy alone versus radiotherapy and chemotherapy.

```
library(mable)  
head(cosmesis, 10)
```

	left	right	treat
1	45	NA	RT
2	6	10	RT
3	0	7	RT
4	46	NA	RT
5	46	NA	RT
6	7	16	RT
7	17	NA	RT
8	7	14	RT
9	37	44	RT
10	0	8	RT

Note that these data include left-censoring, interval-censoring, and right-censoring).



Using the `Surv` function to specify censoring requires that lower bounds of 0 and upper bounds of ∞ be replaced with NA.

```
cosmesis$left <- ifelse(cosmesis$left == 0, NA, cosmesis$left)
head(cosmesis, 10)
```

	left	right	treat
1	45	NA	RT
2	6	10	RT
3	NA	7	RT
4	46	NA	RT
5	46	NA	RT
6	7	16	RT

```

7    17    NA    RT
8     7    14    RT
9    37    44    RT
10   NA     8    RT

```

```
tail(cosmesis, 10)
```

```

      left right treat
85    14    19    RCT
86     4     8    RCT
87    34    NA    RCT
88    30    36    RCT
89    18    24    RCT
90    16    60    RCT
91    35    39    RCT
92    21    NA    RCT
93    11    20    RCT
94    48    NA    RCT

```

It is also useful to note that you can accommodate an observation that is *not* censored by specifying *equal* left and right interval endpoints.

We can verify the censoring specification by looking at what `Surv` produces.

```
cosmesis$y <- with(cosmesis, Surv(left, right, type = "interval2"))
head(cosmesis, 10)
```

```

      left right treat      y
1     45    NA    RT    45+
2      6    10    RT [ 6, 10]
3     NA     7    RT     7-
4     46    NA    RT    46+
5     46    NA    RT    46+
6      7    16    RT [ 7, 16]
7     17    NA    RT    17+
8      7    14    RT [ 7, 14]
9     37    44    RT [37, 44]
10    NA     8    RT     8-

```

Now we can estimate an AFT model.

```

m <- survreg(Surv(left, right, type = "interval2") ~ treat,
  dist = "lognormal", data = cosmesis)
summary(m)$table

```

```

              Value Std. Error      z      p
(Intercept)  3.5479     0.1542 23.013 3.450e-117
treatRCT     -0.4210     0.2032 -2.072 3.827e-02
Log(scale)   -0.1254     0.1095 -1.145 2.521e-01

```

Applying the exponential function helps interpret the effect of the treatment.

```
exp(cbind(coef(m), confint(m)))
```

```

              2.5 %  97.5 %
(Intercept) 34.7394 25.6800 46.9949
treatRCT     0.6564 0.4408 0.9775

```

Using `flexsurvreg` produces the same information but in one output.

```
library(flexsurv)
m <- flexsurvreg(Surv(left, right, type = "interval2") ~ treat,
  dist = "lognormal", data = cosmesis)
print(m)
```

Call:

```
flexsurvreg(formula = Surv(left, right, type = "interval2") ~
  treat, data = cosmesis, dist = "lognormal")
```

Estimates:

	data mean	est	L95%	U95%	se	exp(est)	L95%	U95%
meanlog	NA	3.5479	3.2457	3.8500	0.1542	NA	NA	NA
sdlog	NA	0.8821	0.7118	1.0933	0.0966	NA	NA	NA
treatRCT	0.5106	-0.4210	-0.8192	-0.0228	0.2032	0.6564	0.4408	0.9775

N = 94, Events: 0, Censored: 94

Total time at risk: 2089

Log-likelihood = -146.6, df = 3

AIC = 299.2

Again, it is sometimes helpful for interpretation to change the reference level when dealing with categorical explanatory variables.

```
cosmesis$treat <- relevel(cosmesis$treat, ref = "RCT")
m <- flexsurvreg(Surv(left, right, type = "interval2") ~ treat,
  dist = "lognormal", data = cosmesis)
print(m)
```

Call:

```
flexsurvreg(formula = Surv(left, right, type = "interval2") ~
  treat, data = cosmesis, dist = "lognormal")
```

Estimates:

	data mean	est	L95%	U95%	se	exp(est)	L95%	U95%
meanlog	NA	3.1269	2.8558	3.3980	0.1383	NA	NA	NA
sdlog	NA	0.8821	0.7118	1.0933	0.0966	NA	NA	NA
treatRT	0.4894	0.4210	0.0228	0.8192	0.2032	1.5235	1.0230	2.2688

N = 94, Events: 0, Censored: 94

Total time at risk: 2089

Log-likelihood = -146.6, df = 3

AIC = 299.2

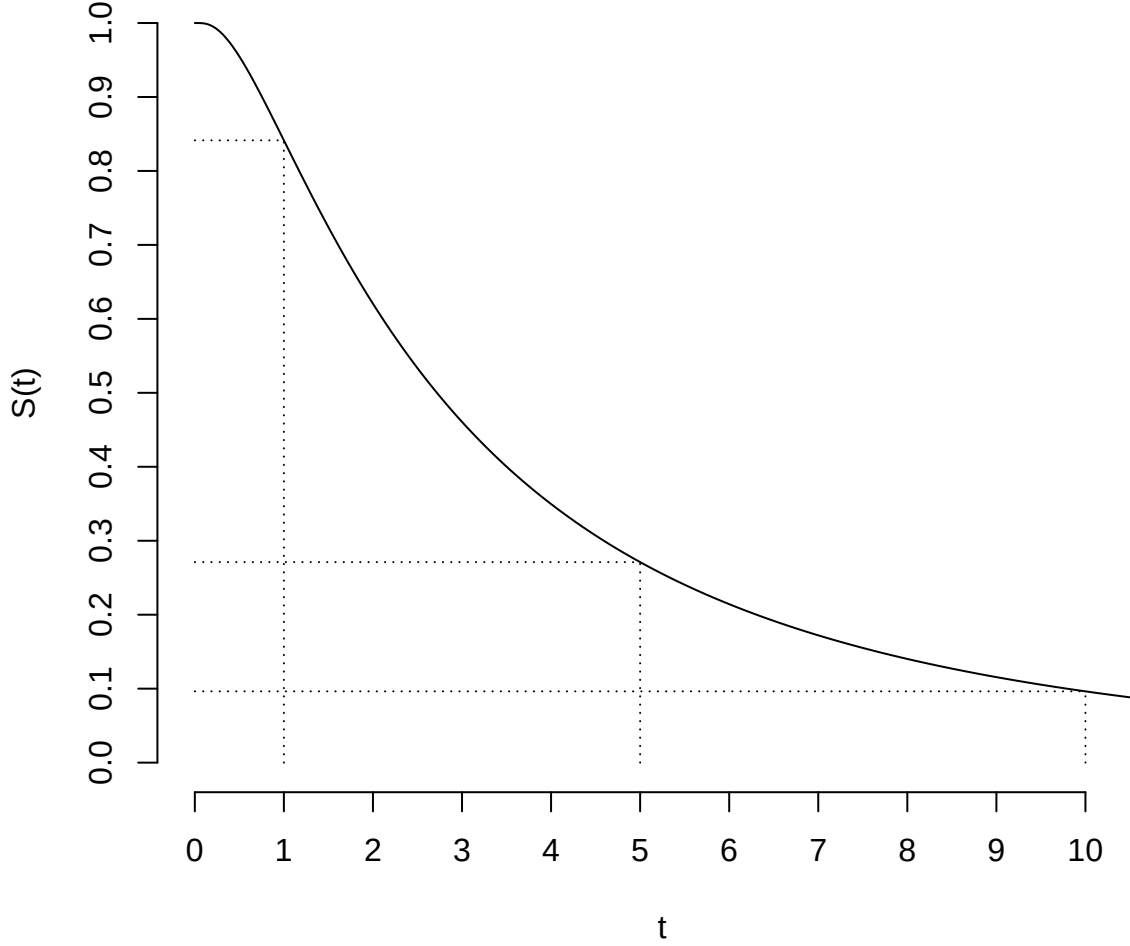
Survival Functions

The *survival function* is

$$S(t) = P(T \geq t),$$

i.e., the probability of a survival time of *at least* t . It is sometimes defined as $S(t) = P(T > t)$ rather than $S(t) = P(T \geq t)$, but if time is modeled as a continuous random variable this distinction does not matter.

Survival Function



Another useful property of survival functions is that the area under the survival curve equals the expected survival time $E(T_i)$, assuming $S(0) = 0$ (i.e., no events have happened at time zero) and $S(\infty) = 1$ (i.e., events eventually do happen).

Survival Functions and AFT Models

Technical Explanation: Accelerated failure time models can be interpreted in terms of effects on survival functions. Let

$$T_b = e^{\beta_0} e^{\beta_1 x_1} e^{\beta_2 x_2} \dots e^{\beta_k x_k} e^{\sigma \epsilon},$$

and let $T_a = e^{\beta_1} T_b$ as before where T_a and T_b are the survival times when the first explanatory variable assumes values of x_a and x_b , respectively. The survival functions for T_a and T_b are then

$$S_a(t) = P(T_a \geq t) \quad \text{and} \quad S_b(t) = P(T_b \geq t),$$

respectively. These survival functions are related because

$$S_b(t) = P(T_b \geq t) = P(e^{\beta_1} T_b \geq e^{\beta_1} t) = P(T_a \geq e^{\beta_1} t) = S_a(e^{\beta_1} t).$$

That is, $S_b(t) = S_a(e^{\beta_1} t)$ and also $S_b(t/e^{\beta_1}) = S_a(t)$. So we can say the following.

1. The probability of survival past t at x_b equals the probability of survival past $e^{\beta_1} t$ at x_a .
2. The probability of survival past t at x_a equals the probability of survival past t/e^{β_1} at x_b .

It can also be shown that we can “order” the survival functions/probabilities from an AFT model because

$$\begin{aligned}\beta_j > 0 &\Leftrightarrow e^{\beta_j} > 1 \Leftrightarrow S_b(t) < S_a(t), \\ \beta_j < 0 &\Leftrightarrow e^{\beta_j} < 1 \Leftrightarrow S_b(t) > S_a(t).\end{aligned}$$

Note that with an AFT model the survival functions at two different values of an explanatory variable do not cross.

In an AFT model the explanatory variables can be viewed as “compressing” or “stretching” time which has the effect of “horizontally compressing/stretching” the survival function. Assume $T_i = e^{\beta_0} e^{\beta_1 x_{i1}} \dots e^{\beta_k x_{ik}} e^{\sigma \epsilon_i}$ and let $S_i(t)$ be the survival function of T_i . Then

$$S_i(t) = P(T_i \geq t) = P(e^{\beta_0} e^{\beta_1 x_{i1}} \dots e^{\beta_k x_{ik}} e^{\sigma \epsilon_i} \geq t) = P[e^{\beta_0} e^{\sigma \epsilon_i} \geq t / (e^{\beta_1 x_{i1}} \dots e^{\beta_k x_{ik}})].$$

If all $x_{ij} = 0$ then $T_i = e^{\beta_0} e^{\sigma \epsilon_i}$ with a “baseline” survival function $S_0(t) = P(e^{\beta_0} e^{\sigma \epsilon_i} \geq t)$. Then

$$S_i(t) = S_0[t / (e^{\beta_1 x_{i1}} \dots e^{\beta_k x_{ik}})] \quad \text{and} \quad S_i(t e^{\beta_1 x_{i1}} \dots e^{\beta_k x_{ik}}) = S_0(t).$$

So the explanatory variables effectively “horizontally” compress or stretch a (hypothetical) baseline survival function. Also in terms of the actual times, if $T_0 = e^{\beta_0} e^{\sigma \epsilon}$ represents a “baseline” survival time when all $x_{ij} = 0$, the

$$T_i = e^{\beta_1 x_{i1}} \dots e^{\beta_k x_{ik}} T_0,$$

so that again the values of the explanatory variables have the effect of “stretching” or “compressing” time time scale.

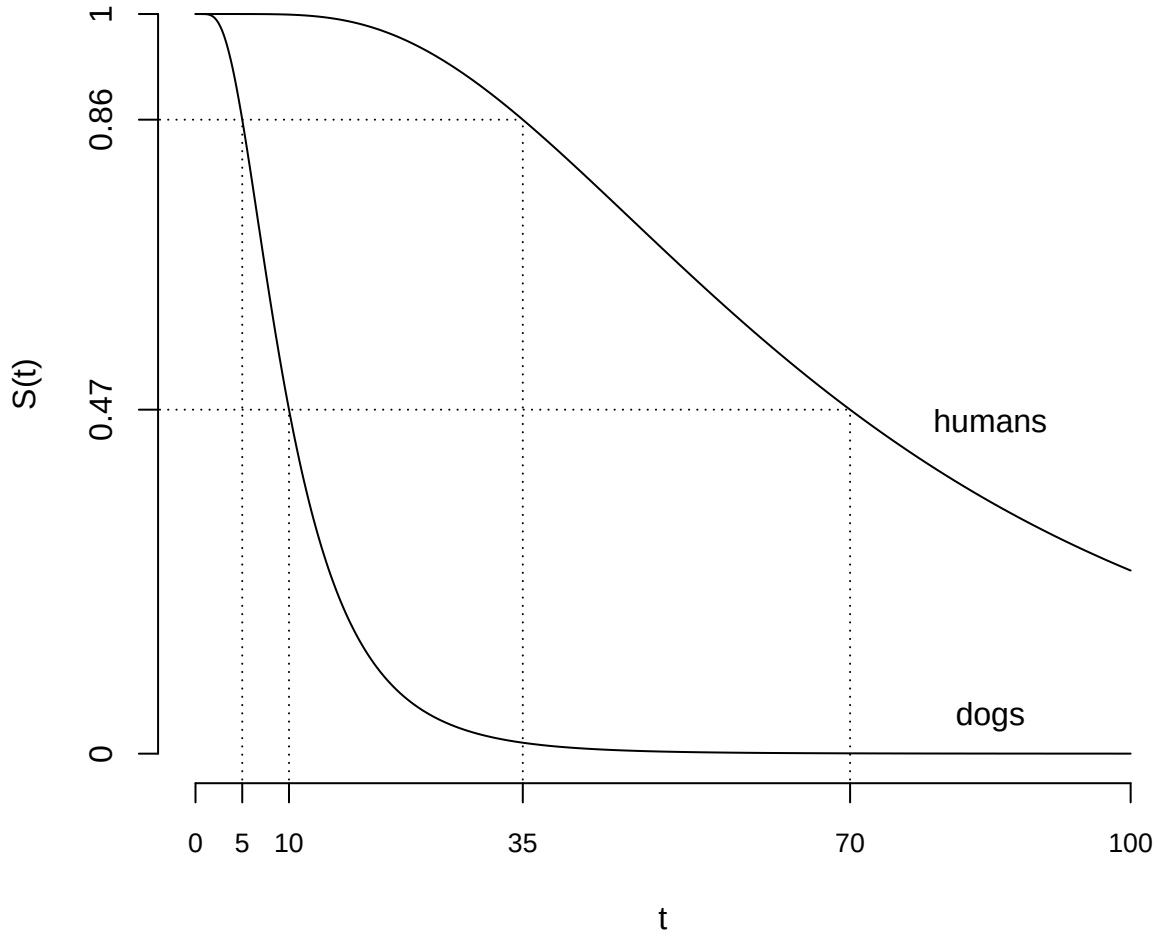
Example: Recall the AFT model for the `lifespan` data where the model is

$$\log T_i = \beta_0 + \beta_1 x_i,$$

where x_i is an indicator variable such that $x_i = 1$ if the species is human (so $x_a = 1$ in the above discussion), and $x_i = 0$ if the species is dog (so $x_b = 0$ in the above discussion). The estimate of β_1 was $\hat{\beta}_1 \approx 1.946$ so that $e^{\hat{\beta}_1} \approx 7$. The “baseline” survival function is the survival function for dogs, which we can write as $S_d(t)$. The survival function for humans is then $S_h(7t)$. For example, we estimate that the probability that a dog lives for 10 or more years equals the probability that a human will live for 70 or more years because $S_d(t) = S_h(7t)$ where $t = 10$. The survival function of a human is obtained by “stretching” the survival function of a dog by a factor of 7.

If we re-parameters the model so that $x_i = 1$ if the species is *dog*, then we have that $\hat{\beta}_1 \approx 1/7$, so that the “baseline” survival function is for humans, and we have that $S_h(t) = S_d(t/7)$. We can also say that we estimate that the probability that a human lives to be 35 or more equals the probability that a dog lives to be 5 or more because $S_d(t/7) = S_h(t)$ where $t = 35$. The survival function of a dog is obtained by “compressing” the survival function of a human by a factor of 1/7.

Survival Functions for Humans and Dogs

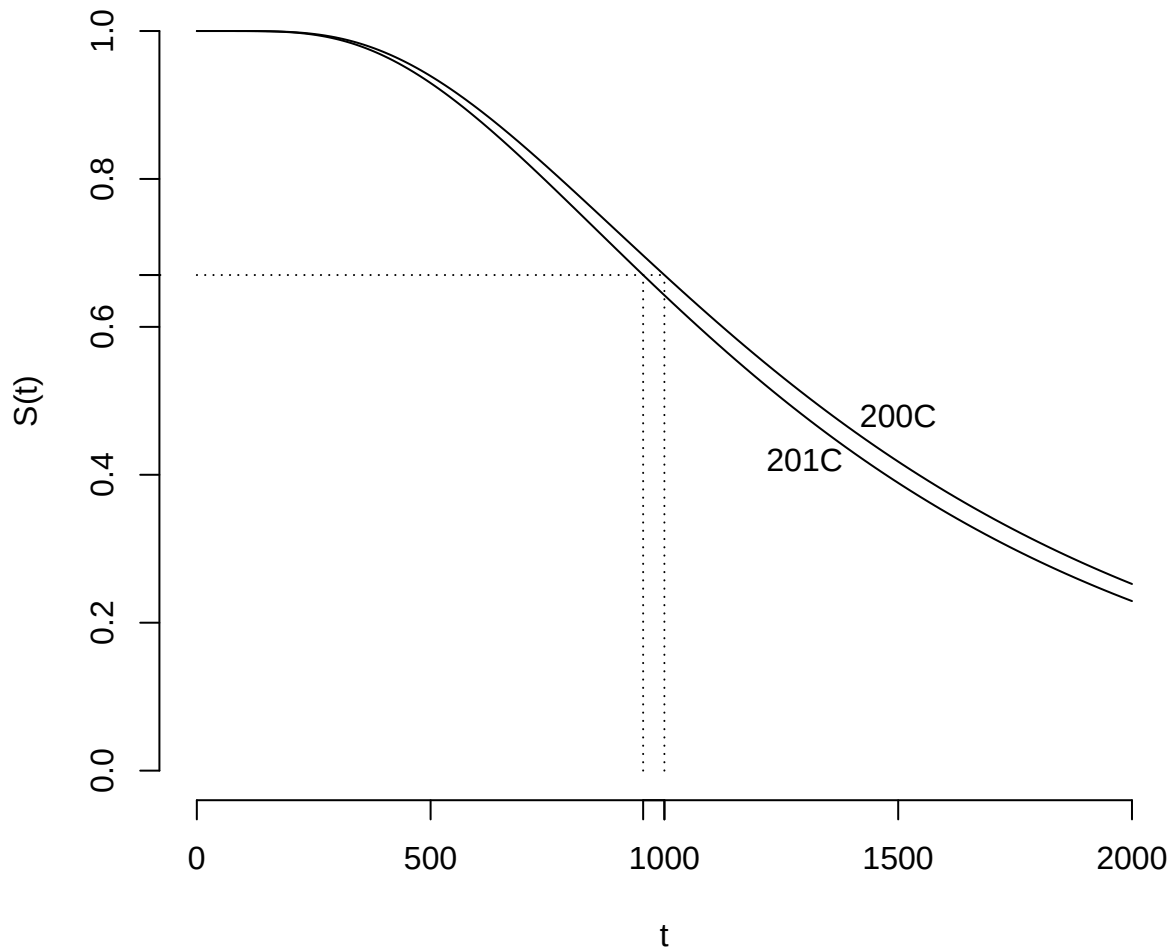


Example: Recall the AFT model for the `motors` data where the model is

$$\log T_i = \beta_0 + \beta_1 x_i,$$

where x_i is temperature. The estimate of β_1 was $\hat{\beta}_1 \approx -0.047$ so that $e^{\hat{\beta}_1} \approx 0.95$. Thus $S_{x+1}(0.95t) = S_x(t)$ where the subscript of x represents temperature. Increasing by one degree “compresses” the survival function by a factor of about 0.95 (i.e., 5%).

Survival Functions at Two Temperatures



Plotting Estimated Survival Functions

Estimating and plotting survival functions is relatively easy using `flexsurvreg` objects. Here the `summary` function behaves more like `predict` for other model objects produced by `lm`, `nls`, and `glm`.

Example: The estimated survival functions for the AFT model for the `lifespan` data can be computed/plotted as follows.

```
library(trtools) # for lifespan data
m <- flexsurvreg(Surv(years) ~ species, dist = "lognormal", data = lifespan)

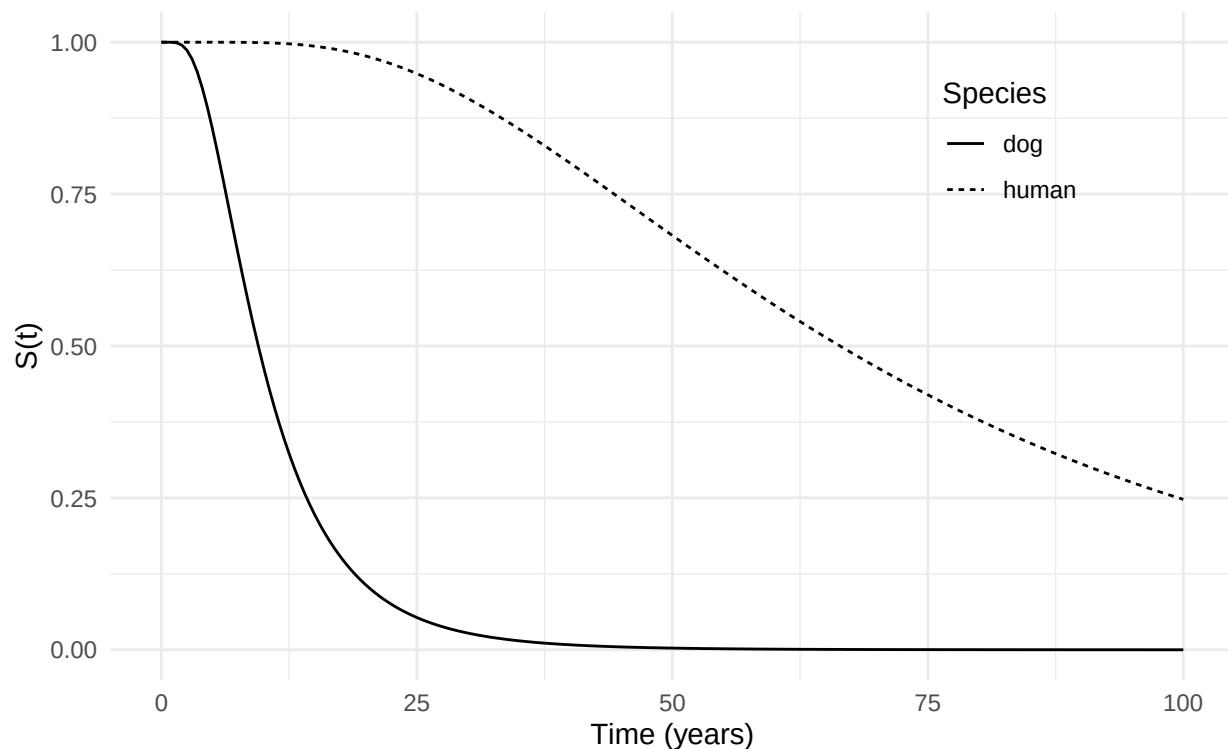
d <- data.frame(species = c("dog", "human"))
d <- summary(m, newdata = d, t = seq(0, 100, by = 0.5), type = "survival", tidy = TRUE)
head(d)
```

	time	est	lcl	ucl	species
1	0.0	1.0000	1.0000	1.0000	dog
2	0.5	1.0000	1.0000	1.0000	dog
3	1.0	0.9999	0.9999	0.9999	dog
4	1.5	0.9989	0.9985	0.9993	dog
5	2.0	0.9953	0.9937	0.9965	dog
6	2.5	0.9869	0.9834	0.9897	dog

```
tail(d)
```

	time	est	lcl	ucl	species
397	97.5	0.2611	0.2412	0.2828	human
398	98.0	0.2583	0.2386	0.2799	human
399	98.5	0.2556	0.2359	0.2771	human
400	99.0	0.2529	0.2333	0.2742	human
401	99.5	0.2502	0.2307	0.2714	human
402	100.0	0.2475	0.2281	0.2686	human

```
p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = species)) +
  labs(x = "Time (years)", y = "S(t)", linetype = "Species") +
  theme_minimal() +
  theme(legend.position = "inside", legend.position.inside = c(0.8,0.8))
plot(p)
```

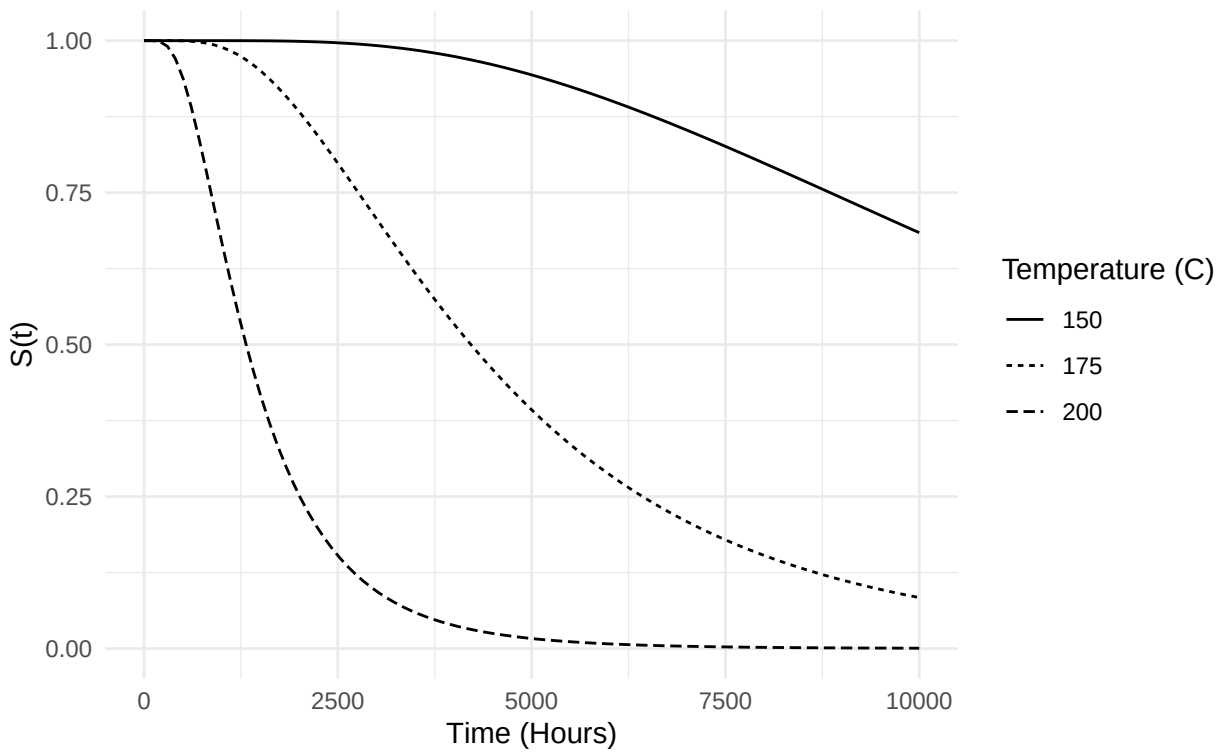


Example: Survival functions at different temperatures based on the AFT model for the `motors` data can be computed/plotted as follows.

```
library(MASS) # for motors data frame
m <- flexsurvreg(Surv(time, cens) ~ temp, dist = "lognormal", data = motors)

d <- data.frame(temp = c(150,175,200))
d <- summary(m, newdata = d, t = seq(0, 10000, length = 100),
  type = "survival", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est, linetype = factor(temp))) +
  geom_line() + theme_minimal() +
  labs(x = "Time (Hours)", y = "S(t)", linetype = "Temperature (C)")
plot(p)
```



```
p <- ggplot(d, aes(x = time, y = est)) +
  geom_line() + facet_wrap(~ temp, nrow = 1) +
  geom_ribbon(aes(ymin = lcl, ymax = ucl), alpha = 0.1) +
  labs(x = "Time (Hours)", y = "S(t)") + theme_minimal()
plot(p)
```

